Group C & D

C: Monte Carlo 101

a) Compile and run the program as is and make sure there are no errors.
   - 1 factor MC with explicit Euler
   - Number of subintervals in time: 100
   - Number of simulations: 50000
   - 10000
   - 20000
   - 30000
   - 40000
   - 50000
   - Price, after discounting: 5.87749,
   - Number of times origin is hit: 0

b) Run the MC program again with data from Batches 1 and 2. Experiment with different value of NT (timesteps) and NSIM (simulations or draws).
   - In particular, how many time steps and draws do you need in order to get the same accuracy as the exact solution?
     1) Monte Carlo does not match the exact Black-Scholes price exactly, because it is based on simulation runs, so the result is slightly different each time, even with the same NT and NSIM. However, I found a general trend: the error is reduced mainly when NSIM increases, while changing NT has a smaller effect in my tested range.
   - How is the accuracy affected by different values for NT/NSIM?
     1) When I tested different NSIM values while keeping NT fixed, I observed a much larger change in error compared with testing different NT values while keeping NSIM fixed. As a result, in this experiment, NSIM was the dominant driver of accuracy, while NT had a secondary impact over the tested range. The reason is that Monte Carlo sampling error decreases at a rate 1/sqrt(NSIM), so increasing NSIM generally improves accuracy more consistently.

c) Now we do some stress-testing of the MC method. Take Batch 4.
   - What values do we need to assign to NT and NSIM in order to get an accuracy to two places behind the decimal point?
     1) To achieve accuracy to two decimal places, the pricing error must be less than 0.005. Based on the stress-testing grid in Batch 4, none of the tested combinations of NT and NSIM produced an error below 0.005 for both the call and the put. The closest case was NT = 1000 and NSIM = 200000, where the call error was 0.00485 (meeting the requirement) but the put error was 0.01331, which is still above the threshold. Therefore, the tested grid suggests that a greater computational effort is needed, such as increasing the number of simulations further or performing repeated runs and averaging the results, in order to reliably obtain two-decimal accuracy for both option prices.

- How is the accuracy affected by different values for NT/NSIM?
    1) The accuracy of the Monte Carlo method is influenced by two factors: sampling error and discretization error. The number of simulations NSIM mainly controls the Monte Carlo sampling error; increasing NSIM generally improves accuracy because the estimator converges to the true value, with error decreasing roughly at a rate proportional to 1/sqrt(NSIM). However, due to randomness, the observed error in a single run may not decrease monotonically. The number of time steps NT affects the discretization error of the Euler approximation. Larger NT values reduce this bias by making the simulated paths closer to the continuous time. In this experiment, the long maturity (T=30) amplifies discretization effects, so changes in NT can significantly affect the accuracy of the results.

D: Advanced Monte Carlo

a) Create generic functions to compute the standard deviation and standard error
    - 1 factor MC with explicit Euler
    - Number of subintervals in time: 100
    - Number of simulations: 50000
    - 10000
    - 20000
    - 30000
    - 40000
    - 50000
    - Price, after discounting: 5.87749,
    - Standard deviation: 6.06167
    - Standard error: 0.0271086
    - Number of times origin is hit: 0

b) Run the MC program again with data from Batches 1 and 2. Experiment with different values of NT (time steps) and NSIM (simulations or draws).
    - How do SD and SE react for these different run parameters, and is there any pattern in regards to the accuracy of the MC (when compared to the exact method)?
        1) SD and SE change when the number of simulations NSIM changes while NT remains fixed, because both statistics depend on the number of simulated paths used in the MC estimation. As NSIM increases, the estimate is based on more simulated outcomes, which generally reduces the variability. Specifically, SE is more sensitive to changes in NSIM than SD, since SE = SD/sqrt(NSIM). This means that increasing the number of simulations directly reduces the SE and improves the reliability of the MC estimate. In terms of accuracy, the pattern is that when SD and SE decrease, the MC price tends to be closer to the exact solution, resulting in smaller pricing errors. Therefore, lower SD and SE generally indicate a more stable simulation and higher accuracy of the MC method compared to the exact price.